GlucoTwin | System Architecture

A Personalized Glucose Digital Twin • Digital Twin Challenge 2026 • Individual research prototype

Purpose: explore short-term glucose trend forecasting, potential glucose-event classification, evaluation and interactive visualization using synthetic data.

DATA FOUNDATION

1 SYNTHETIC DATA

`data/generate_synthetic_data.py`

Patient profiles + glucose time series

2 VALIDATION

`validate_data.py`

Data quality checks before modelling

MODEL TRAINING

3A FORECASTING

`train_model.py`

Random Forest regressor • +30 min horizon • latest-20% holdout

10 input features

3B EVENT CLASSIFICATION

`train_event_model.py`

Random Forest classifier • event ≥180 mg/dL within 60 min

Balanced classes

EVALUATION

4A FORECAST EVALUATION

`evaluate_forecast.py`

Baseline comparison against persistence

MAE 8.373 vs 9.278

4B EVENT EVALUATION

`evaluate_events.py`
`analyze_event_thresholds.py`

Confusion matrix, P / R / F1

F1 0.431 @ 0.5

OUTPUT

5 REPORT

`EVALUATION_REPORT.md`

Metrics, trade-offs and limitations from the synthetic evaluation.

402	rows evaluated
18	held-out events
0.8	best-F1 threshold
−9.75%	MAE vs baseline

6 INTERACTIVE APPLICATION

`app.py`

Glucose history | **Forecasts** | **Events** | **Data explorer** | **Model summary**

Streamlit

SAFETY BOUNDARY Synthetic-data research prototype. Not clinically validated. Not for diagnosis, treatment, or medication decisions.